CAPSTONE PROJECT

E-commerce Data Analysis Project

Project Overview

This project demonstrates how data science is used in e-commerce to analyze customer purchasing behavior, sales performance, and business trends. The analysis focuses on identifying relationships between customer demographics, product categories, transaction patterns, and sales outcomes to generate meaningful insights for business decision-making and predictive analytics.

The project uses a structured e-commerce dataset containing customer information, product details, order transactions, purchasing behavior, payment methods, shipping details, and revenue data. Through data cleaning, exploratory data analysis (EDA), visualization, and machine learning techniques, the project aims to uncover patterns that influence customer behavior, sales performance, and business growth.

Project Objectives

The project seeks to:

- Clean and preprocess e-commerce data for analysis.
- Explore sales trends and customer purchasing patterns.
- Investigate how customer demographics influence purchasing behavior.
- Examine the impact of product categories, discounts, and promotions on sales performance.
- Analyze customer spending patterns and purchase frequency.
- Compare sales performance across regions, cities, or online platforms.
- Build predictive machine learning models for sales forecasting or customer purchase prediction.
- Evaluate model performance and identify the most important factors influencing sales and customer behavior.

Dataset Description

The E-commerce Dataset contains structured tabular data describing customer transactions, product information, and purchasing behavior that may influence business performance and revenue generation.

The dataset includes the following variables:

Variable	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer
Location	Geographic location of the customer
Product Category	Type/category of product purchased
Product Price	Price of the purchased product
Quantity	Number of items purchased
Order Date	Date of purchase
Payment Method	Method used for payment

Variable	Description
Discount	Discount applied to the purchase
Shipping Cost	Cost of product delivery
Purchase Amount	Total purchase expenditure
Customer Rating	Customer satisfaction rating
Order Status	Status of order (completed, canceled, returned)

Data Science Workflow

The project follows a standard e-commerce analytics workflow:

1. Data Collection and Loading

Import the e-commerce dataset into Python for analysis.

2. Data Cleaning and Preprocessing

Handle missing values, remove duplicates, process date variables, encode categorical variables, and prepare the data for modeling.

3. Exploratory Data Analysis (EDA)

Explore purchasing behavior, customer spending patterns, and sales trends.

4. Data Visualization

Use charts, histograms, bar plots, heatmaps, and trend lines to visualize customer behavior and business performance.

5. Feature Engineering and Selection

Create useful variables such as purchase frequency, average spending, seasonal sales indicators, and identify important features affecting sales.

6. Machine Learning Model Development

Train predictive models for:

- Sales prediction
- Customer segmentation
- Purchase behavior prediction
- Customer churn analysis
- Revenue forecasting

7. Model Evaluation

Assess prediction accuracy using evaluation metrics such as:

- **Mean Absolute Error (MAE)**
- **Mean Squared Error (MSE)**
- **Root Mean Squared Error (RMSE)**
- **R-squared (R^2)**
- **Accuracy, Precision, Recall, and F1-score** (for classification tasks)

8. Interpretation and Insights

Identify major drivers of customer purchasing behavior and interpret findings for business growth and decision-making.

Expected Outcomes

By completing this project, users will be able to:

- Understand how e-commerce data is analyzed using data science techniques.
- Perform professional exploratory data analysis in Python.
- Analyze customer purchasing behavior and sales trends.
- Build predictive models for sales forecasting and customer behavior prediction.
- Interpret relationships between products, customers, and revenue.
- Apply machine learning techniques in e-commerce analytics.
- Generate actionable insights for improving customer experience and business performance.

Applications of the Project

This project has practical applications in:

- **Customer behavior analysis**
- **Sales forecasting**
- **Recommendation systems**
- **Inventory management**
- **Customer segmentation**
- **Marketing analytics**
- **Business intelligence**
- **Revenue optimization**
- **Data science and machine learning education**

Conclusion

E-commerce analytics is an important application of data science in modern digital business environments. By analyzing customer demographics, purchasing patterns, product preferences, and sales trends, businesses can better understand customer behavior, optimize operations, and improve revenue generation. This project provides a practical framework for applying data science techniques to real-world e-commerce data using Python or R

Dataset: E-commerce dataset

It is suitable for **EDA, machine learning, sales forecasting, customer segmentation, and predictive analytics in Python or R**